

2/16/26

Hw3.

1) Π name (O specialty = 'Cardiology' (Doctors))

Answer: name

Dr. Sarah Chen

Dr. James Kim

2) Π name, age (Patients)

Answer: <u>name</u>	<u>age</u>
John Martinez	45
Emily Wang	32
Robert Johnson	58
Lisa Patel	28

3) Π name (O age < 40 (Patients))

Answer: name

Emily Wang

Lisa Patel

4) Π name (O years_exp > 10 (Doctors))

Answer: name

Dr. Sarah Chen

Dr. Maria Lopez

Dr. James Kim

5) $\Pi_{name} (\sigma_{insurance = 'Blue Cross'} (Patients))$

Answer: name

John Martinez

Robert Johnson

6) $\Pi_{name} (\sigma_{specialty = 'Cardiology', years_exp > 10} (Doctors))$

Answer: name

Dr. Sarah Chen

Dr. Maria Lopez

7) $\Pi_{name, date, status} (Patients \bowtie Appointments)$

Answer

<u>name</u>	<u>date</u>	<u>status</u>
John Martinez	2024-12-15	Completed
Emily Wang	2024-12-16	Completed
Robert Johnson	2024-12-20	Scheduled
John Martinez	2024-12-22	Scheduled
Lisa Patel	2024-12-18	Cancelled

8.) Π Appointments.appt_id, Appointment.date, Doctors.name
 (O status = 'complete' (Appointments) $\bowtie$ Doctors)

Answer:

app_id	date	name
A001	2024-12-15	Dr. Sarah Chen
A002	2024-12-16	Dr. Ahmed Hassan

9.) Π Patients.name (O Doctors.specialty = 'Cardiology' $\bowtie$ Appointments) $\bowtie$ Patients)

Answer: name
 John Martinez
 Robert Johnson

10.) ρD_1 (Sailors)

ρD_2 (Sailors)

Π D1.name, D2.name, D1.specialty (O D1.doc_id < D2.doc_id)

ρD_1 (Doctors) $\bowtie$ D1.specialty = D2.specialty ρD_2 (Doctors)

Answer:

D1 name	D2 name	specialty
Dr. Sarah Chen	Dr. Maria Lopez	Cardiology